

How Does ChatGPT Actually Work?

From Your Question to an AI Answer

Explain why the sky is blue to a 10-year-old.



The sky looks blue because...

INSIDE THE AI



Real Concepts



Clear Explanations



Practical Insights



For Curious Learners

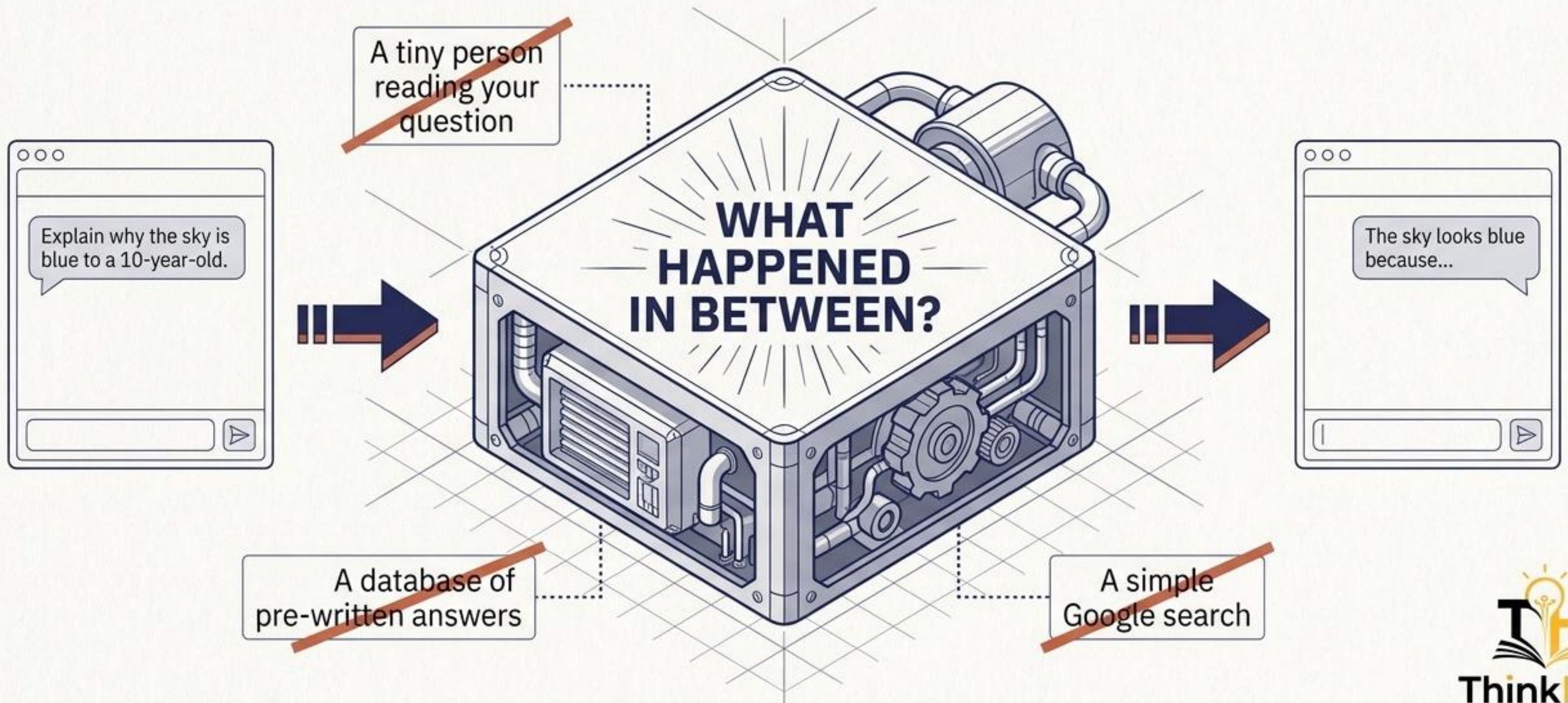
Presented by

Daksh Khandelwal
Founder, ThinkHub

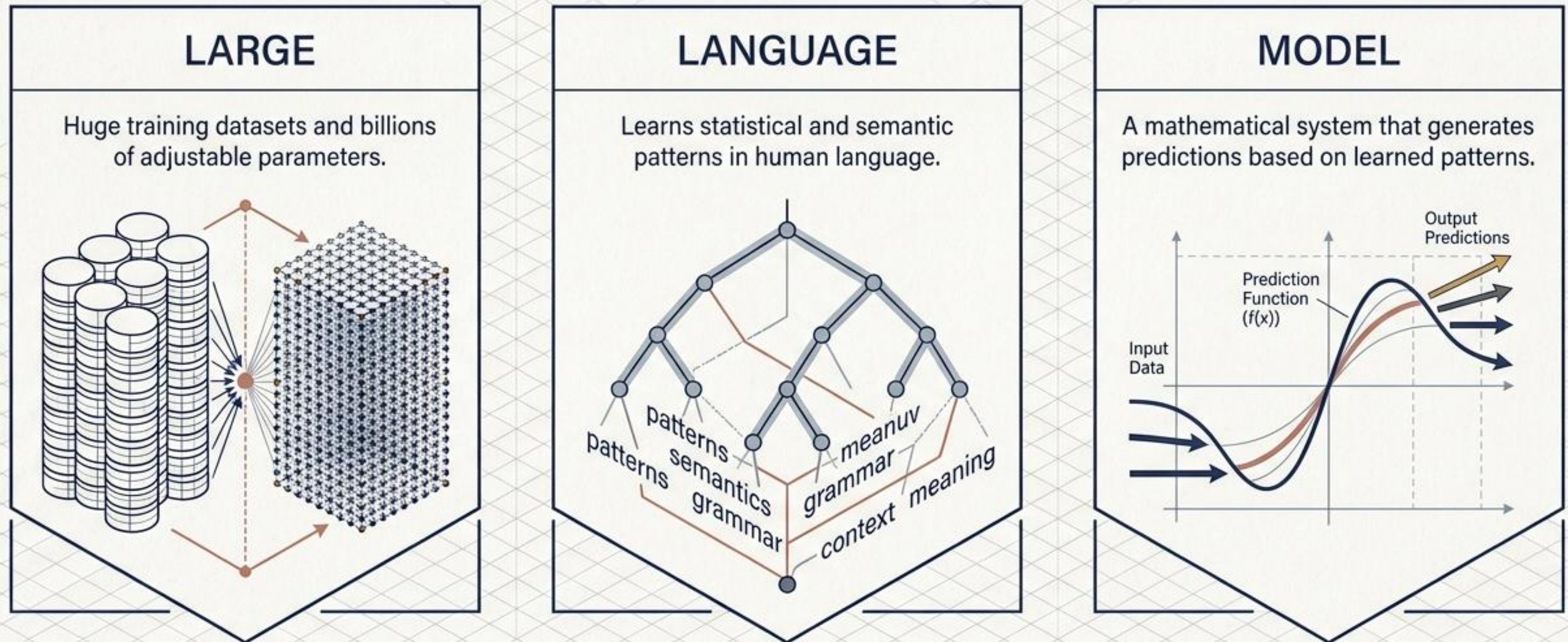
Let's Understand AI Together

What Happens After You Press Send?

Inside the computational system behind modern conversational AI.



The Foundation of Modern AI

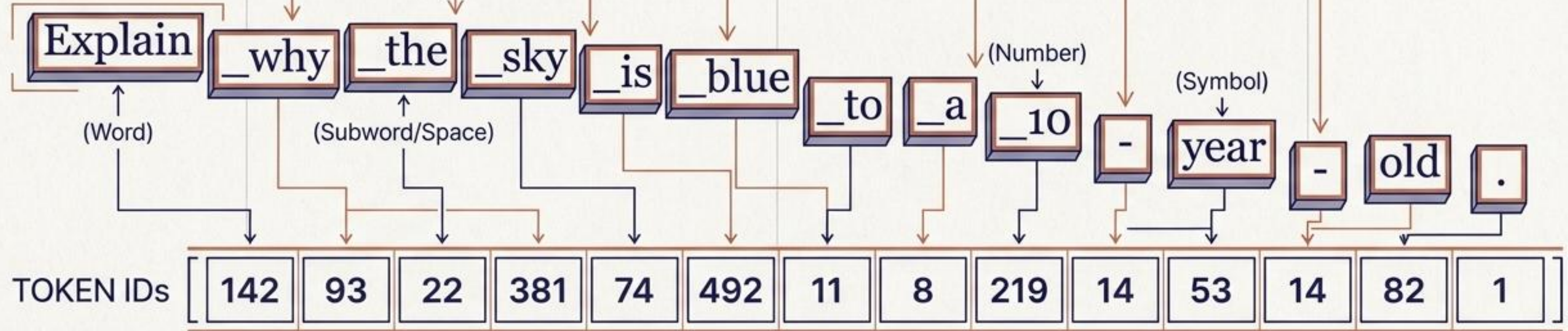


LLM = Large Language Model. A computational system with learned parameters, not a digital brain.

Shattering the Sentence into Tokens

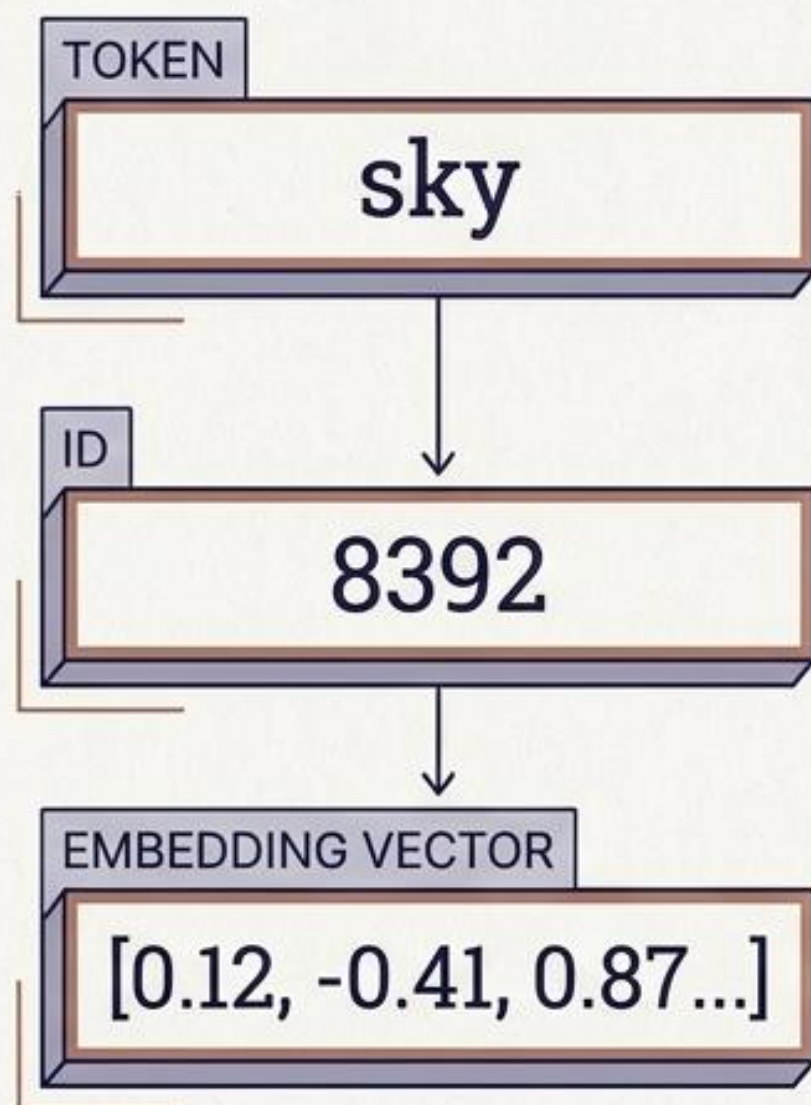
Explain why the sky is blue to a 10-year-old.

TEXT

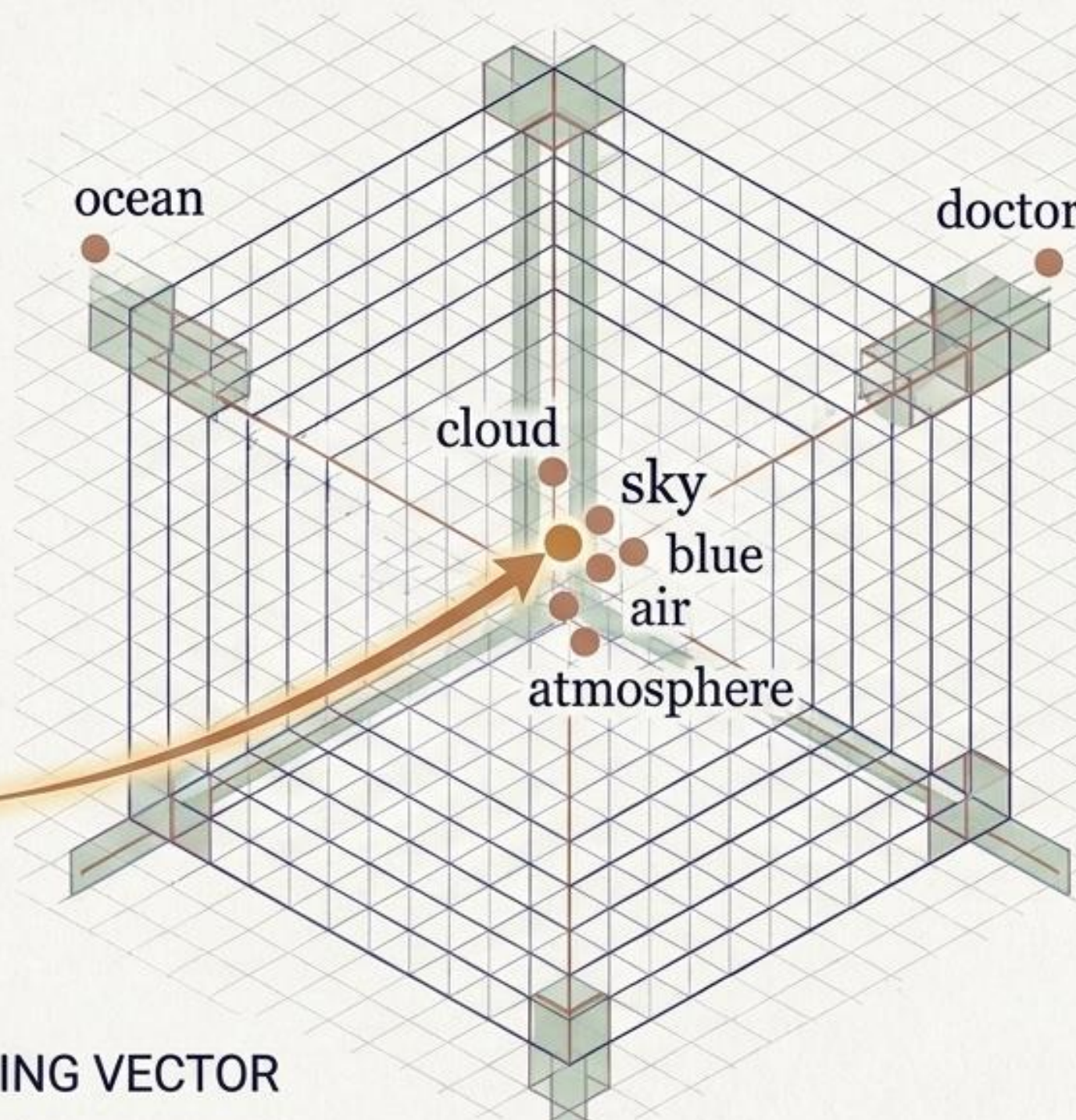


Token $\neq$ Word. Computers operate on numbers.
The tokenizer breaks text into machine-readable pieces.

Mapping Tokens into Mathematical Space



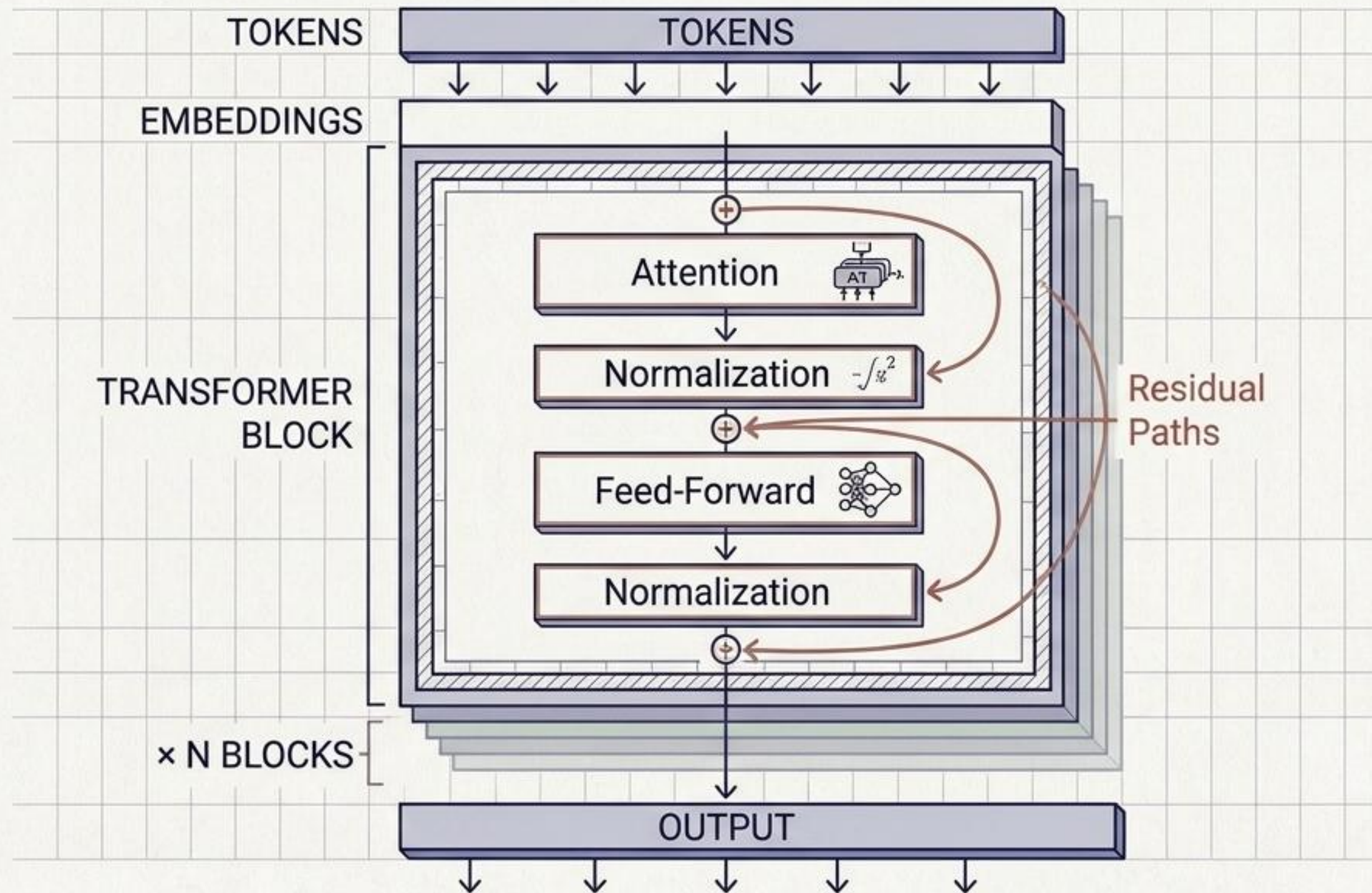
TOKEN → TOKEN ID → EMBEDDING VECTOR



Context changes representations.

A token does not have one permanent, fixed meaning vector; its representation adapts based on surrounding tokens as it passes through the network.

The Architecture That Changed Language AI



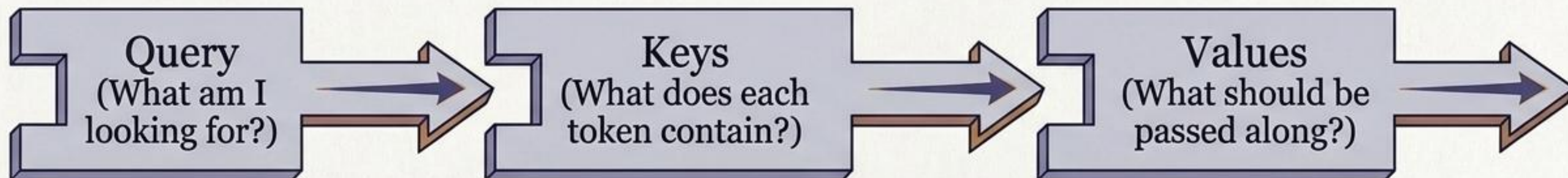
The Transformer moves away from reading text strictly left-to-right. It uses repeated computational blocks to transform representations simultaneously.

Attention Evaluates Contextual Relationships

THE ANIMAL DIDN'T CROSS **it** TIRED THE STREET BECAUSE



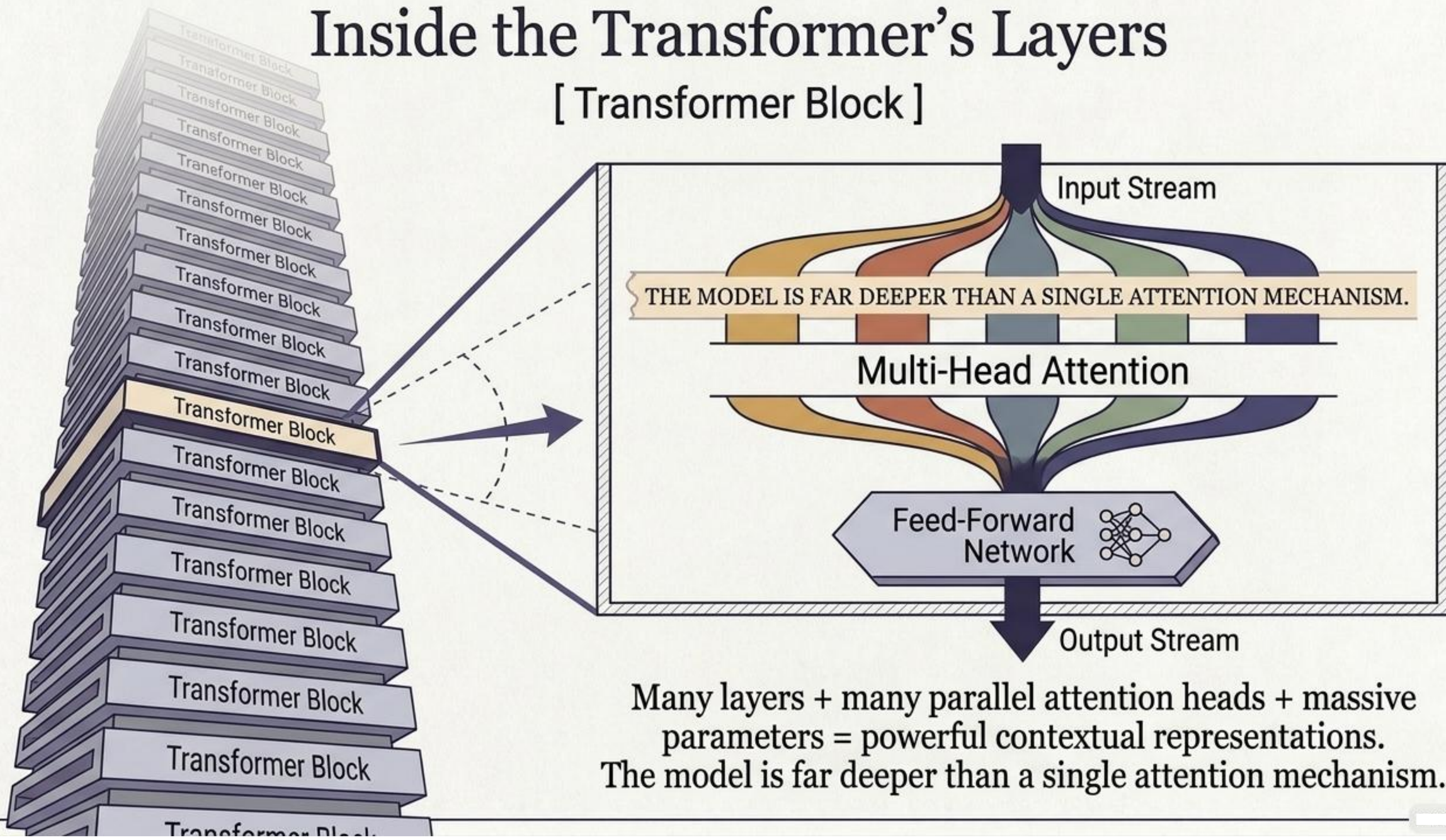
THE CAT SAT ON **it** THE MAT BECAUSE



Attention = Learning which parts of the surrounding context are most relevant to understanding the current token.

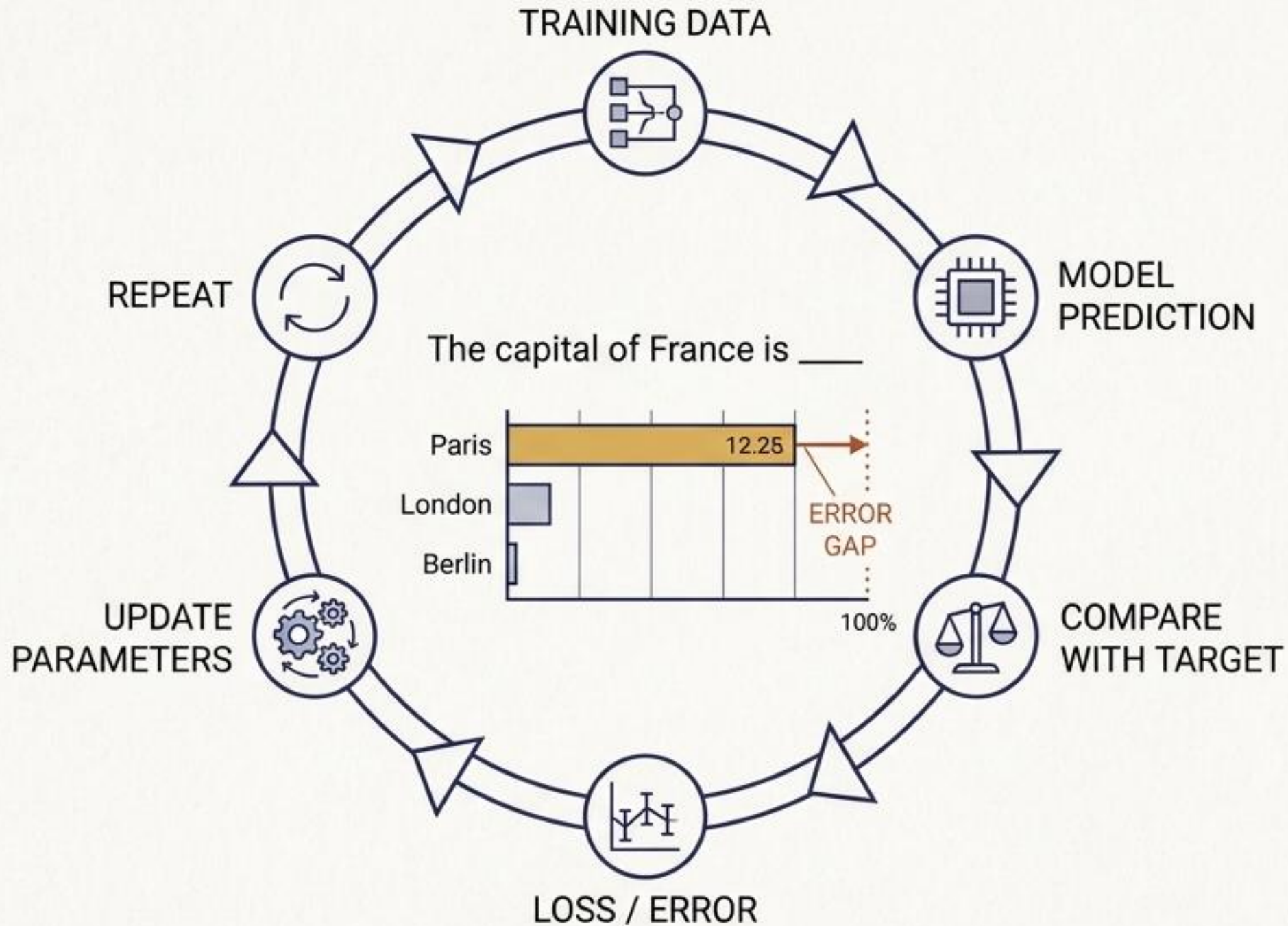
Inside the Transformer's Layers

[Transformer Block]



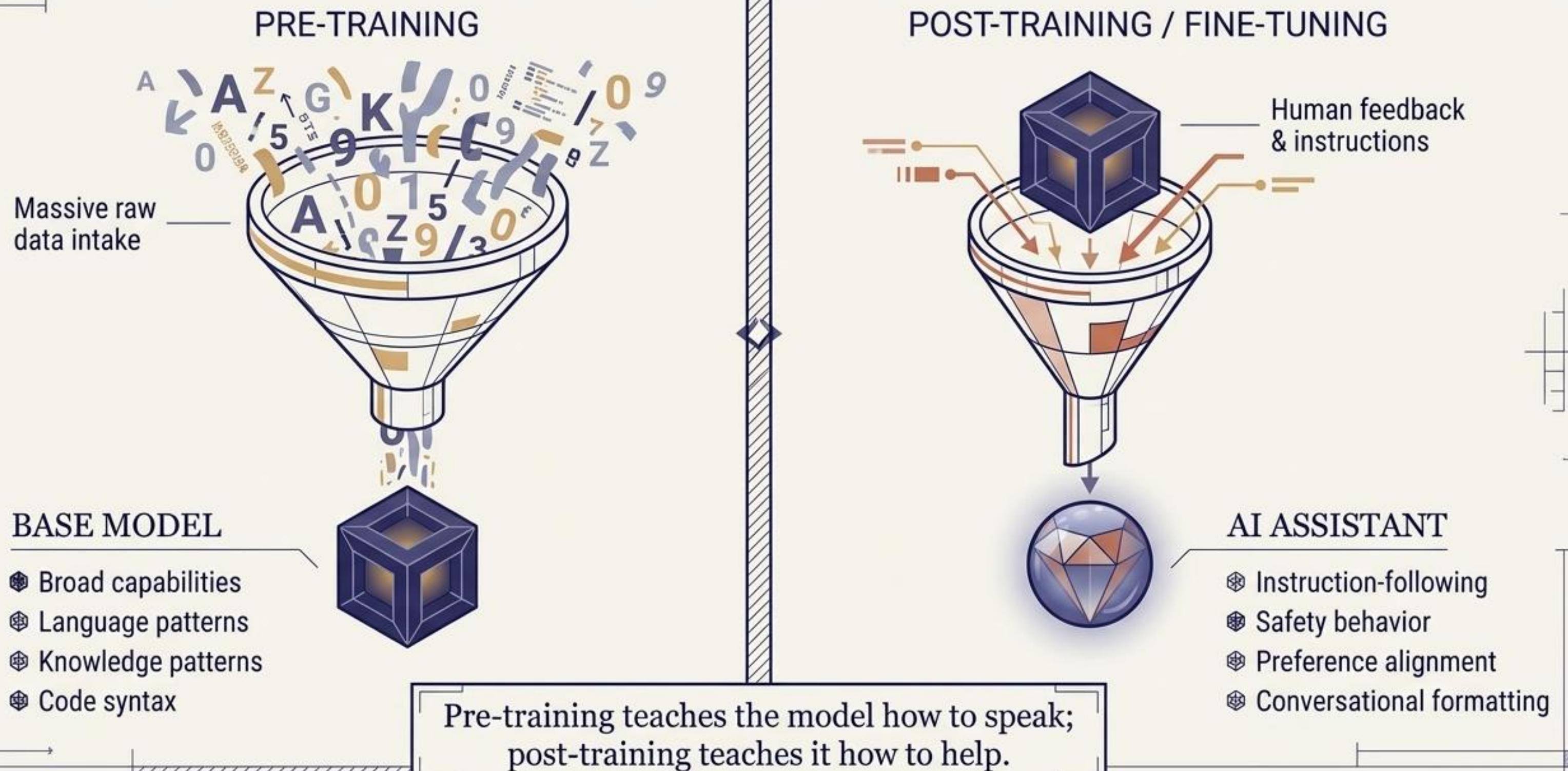
Many layers + many parallel attention heads + massive parameters = powerful contextual representations.
The model is far deeper than a single attention mechanism.

How the Machine Learned to Predict

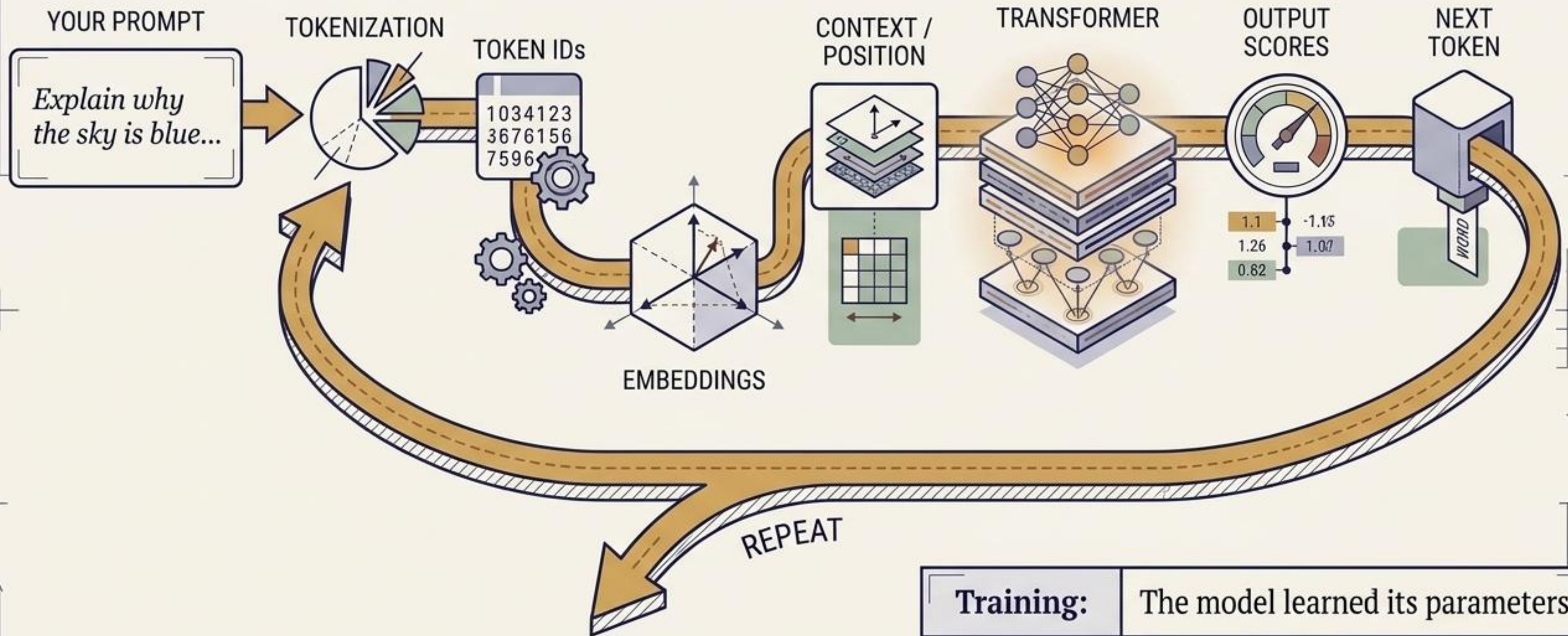


Training is not memorizing a giant answer book. It is adjusting billions of mathematical parameters (knobs) to better predict the next token based on context.

Turning a Raw Model into an Assistant

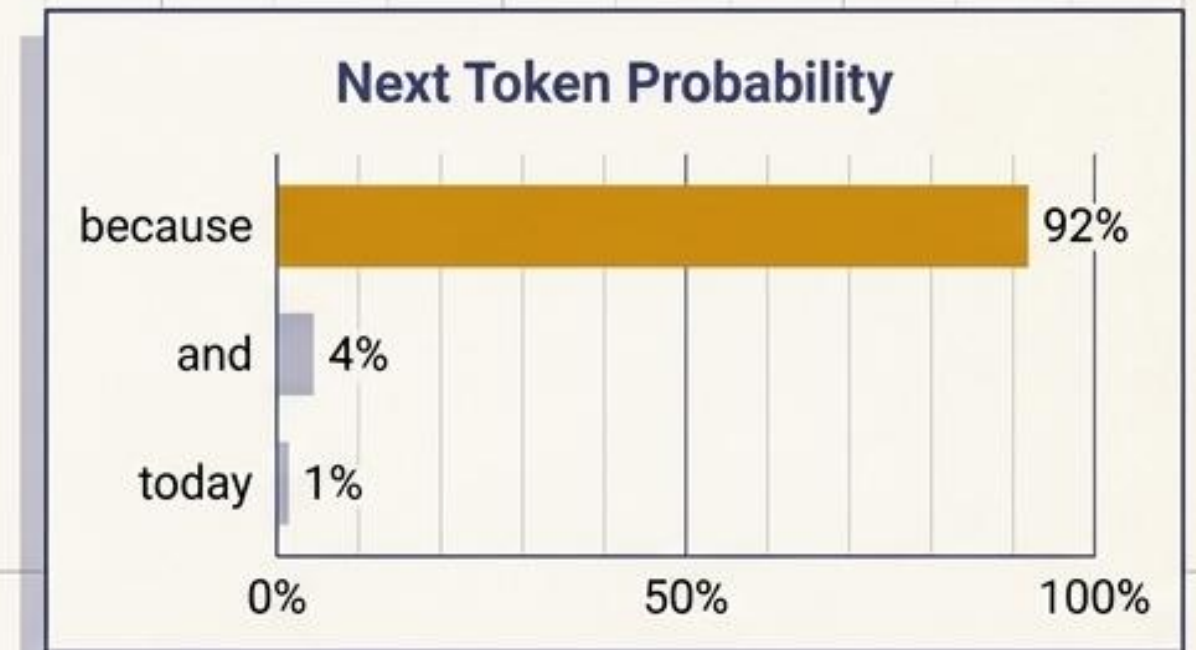
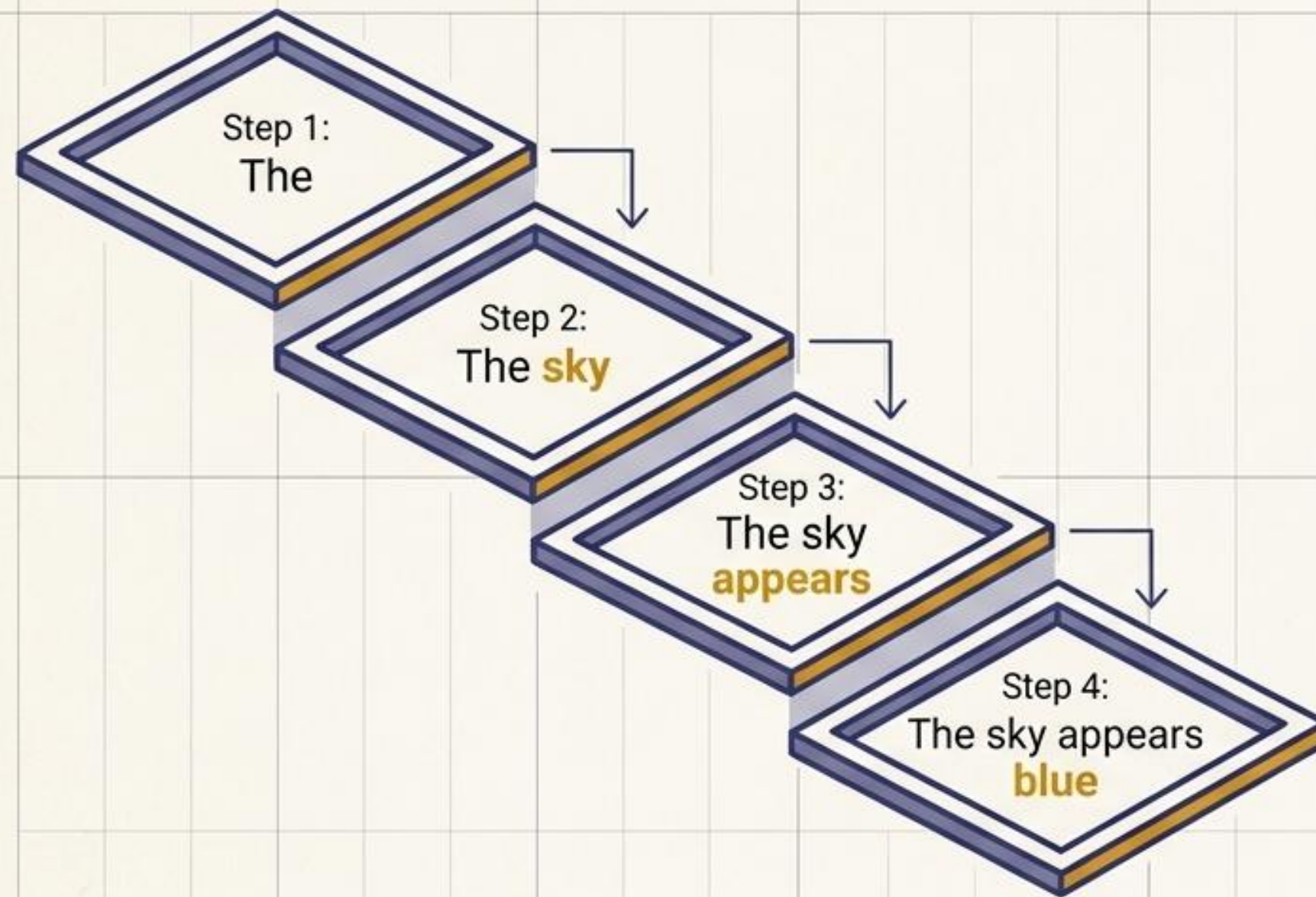


The Moment You Press Send: The Inference Pipeline



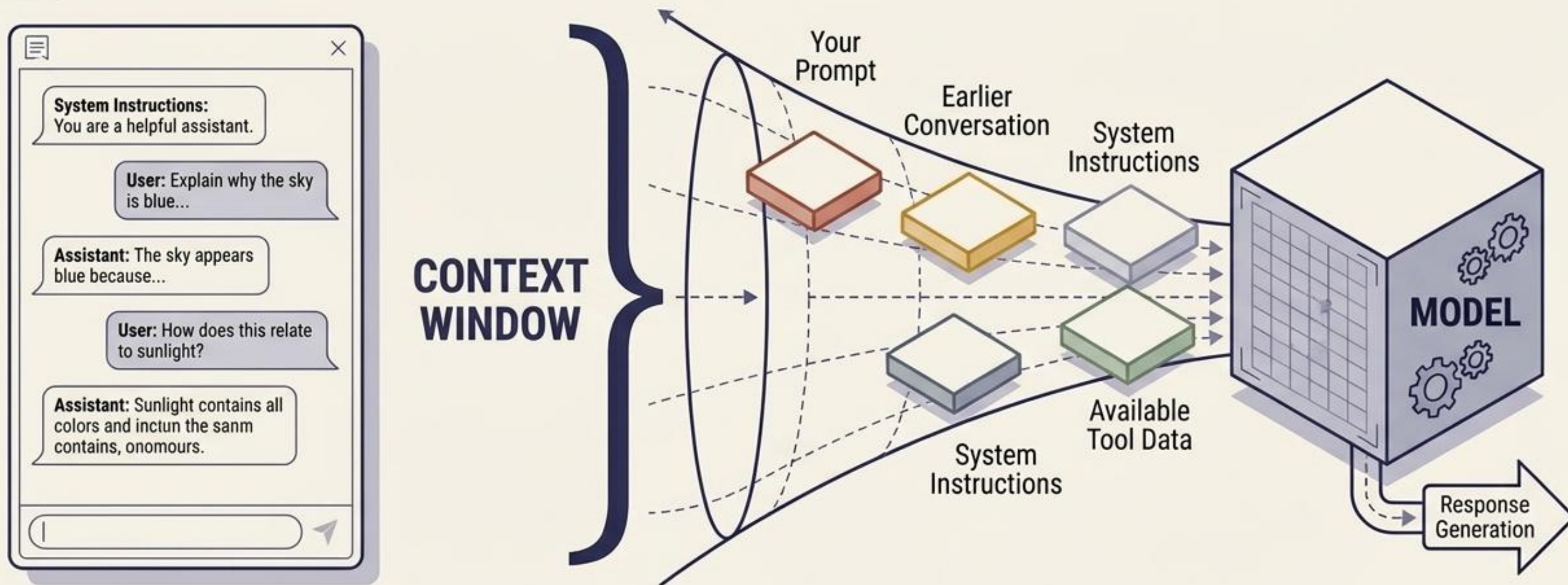
Training:	The model learned its parameters.
Inference:	The trained model is now being used to generate an answer.

Generating the Answer One Token at a Time



Next-token prediction is the core mechanism, but the emergent capabilities are far more sophisticated than simple autocomplete.

The Finite Memory of the Context Window



Context:

Provides temporary information for the current interaction.
(Like holding a document in short-term memory).

Training:

Permanently changes the model's learned parameters.
(Like altering the structure of the brain).

When Fluency Masks Fiction (Hallucinations)

The Illusion

The scientist Dr. Arjun Mehta invented the Quantum Banana Engine in 1987...

FALSE / UNSUPPORTED

The Mechanical Truth

The core objective is generating likely continuations, not guaranteeing truth.

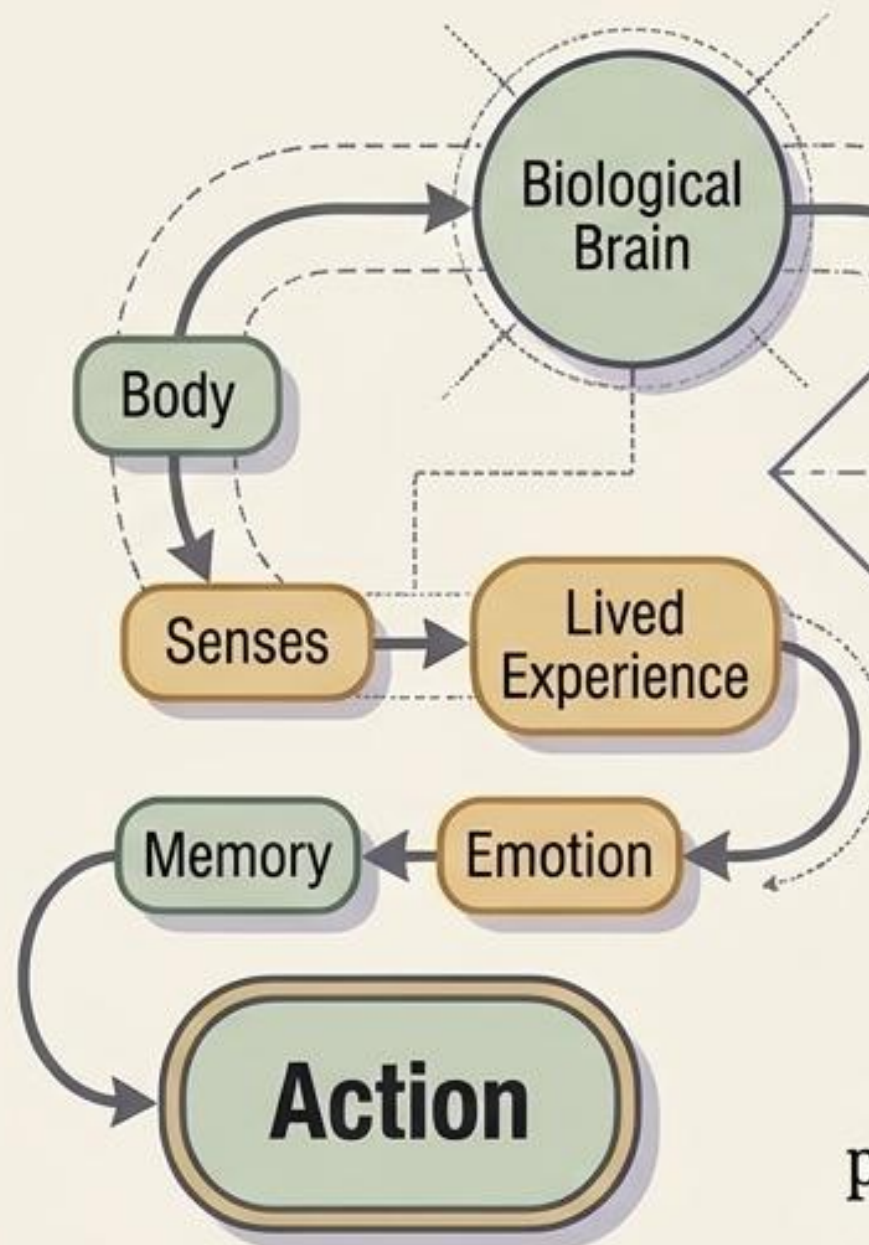
Fluency $\neq$ Truth.

Plausible $\neq$ Accurate.

Confidence in the wording is not evidence that the claim is true.

Computational Processing vs. Human Thought

HUMAN



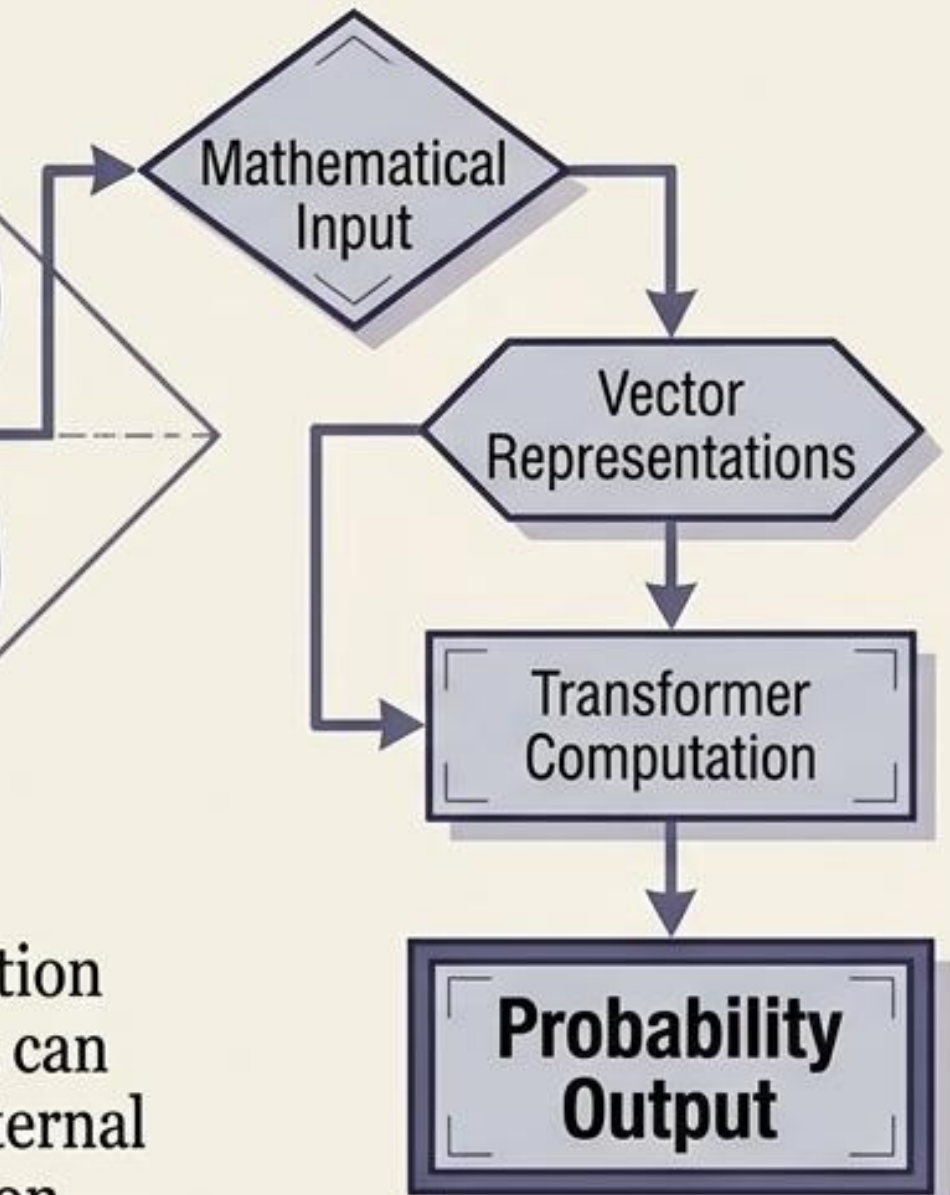
Reasoning-like behavior



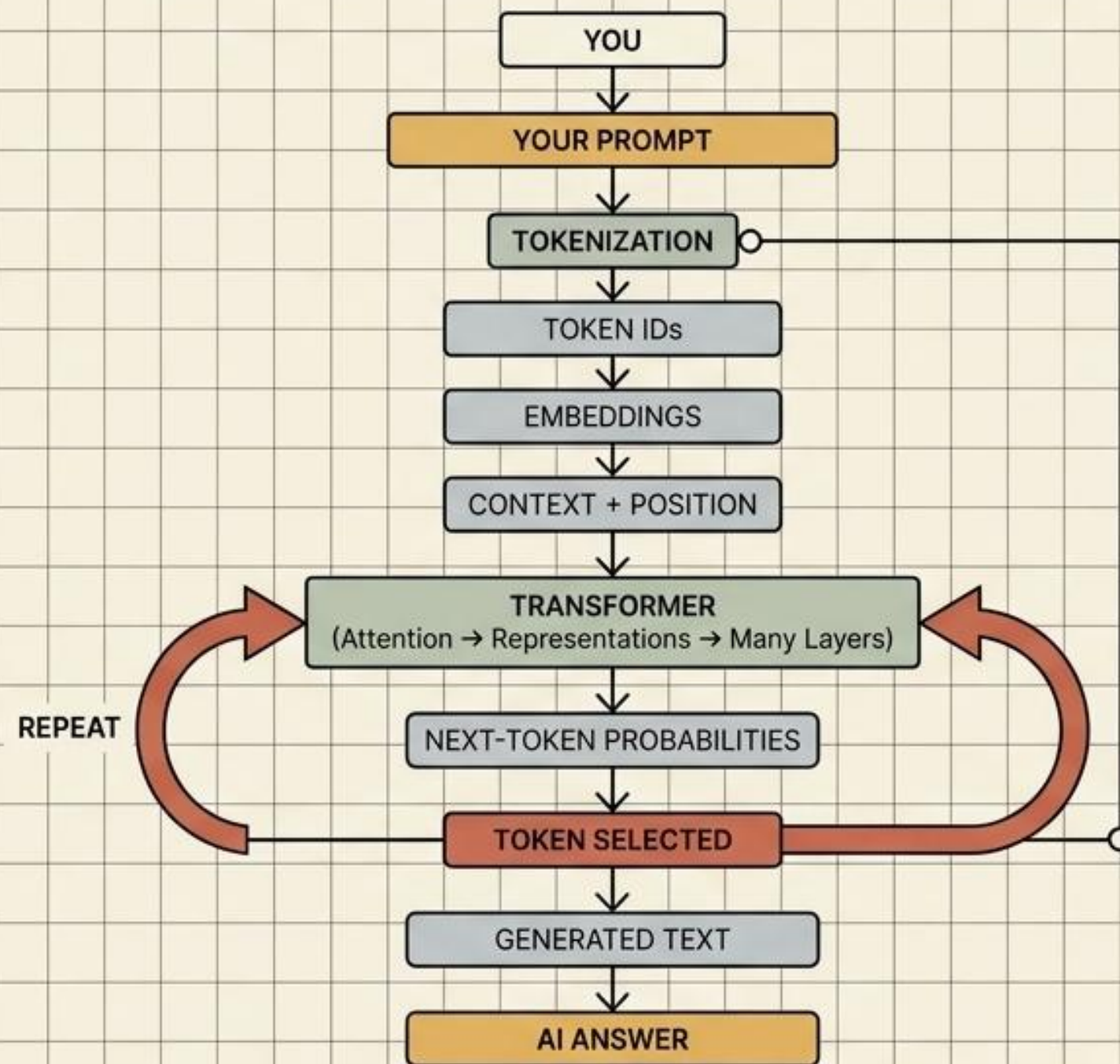
Human consciousness

An LLM performs mathematical computation over representations learned from data. It can produce reasoning-like solutions, but its internal process is not identical to human cognition.

AI SYSTEM



The Complete Picture: From Prompt to Answer



1. Training creates the learned model.

The Training creates the component to create the training model. The current learned model is meta with evaluations will help to learn topics in the models.

2. Inference uses that model to generate tokens.

The Inference uses that model to generate tokens, even model has to generate tokens. Write the answer using model uses that model to generate tokens.

3. Fluent output does not guarantee truth.

Fluent output does not guarantee truth. We cannot guarantee truth. Because of fluent output, the rate that AI answer does not compare to humans.



THANK YOU!

The future belongs to those who **build it continuously.**



START TODAY.
Take the first step.



STAY CONSISTENT.
Small steps, big results.



BUILD ONE PROJECT
at a time.



PRESENTED BY
DAKSH KHADELWAL
FOUNDER, THINKHUB



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